

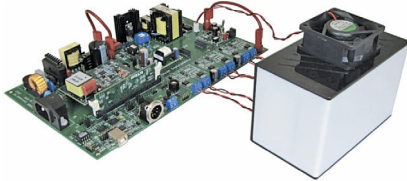
DALI development, accelerates testing, and helps you focus on building reliable DALI-based lighting products.

Link. <https://www.electronicsforu.com/electronics-projects/dali-2-lighting-system-reference-design>

OEM Brand. Renesas

AC-mains LED lighting with DALI

DALI-based LED lighting systems require stable power delivery, smooth dimming, and reliable two-way communication, while also meeting efficiency and AC compliance requirements. This reference design simplifies development by



combining an AC-input LED driver with built-in DALI support, allowing lighting fixtures to be dimmed, addressed, and controlled remotely without additional communication hardware. The design supports multiple LED strings with independent dimming, maintains stable LED current during DALI commands, and delivers high efficiency and power factor. By integrating power conversion, dimming control, and DALI communication into a single platform, it reduces development time, lowers design risk, and simplifies the creation of networked DALI lighting systems for real-world installations.

Applications. A wide range of DALI-enabled LED lighting applications.

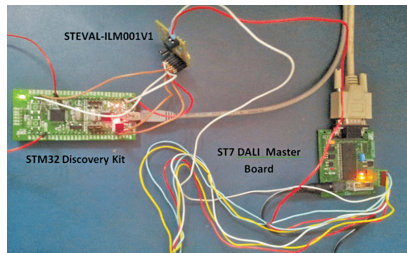
Link. <https://www.electronicsforu.com/electronics-projects/ac-mains-led-lighting-with-dali-reference-design>

OEM Brand. Texas Instruments (TI)

Smarter lighting with a DALI slave

This reference design demonstrates

how you can build a DALI slave device for smart lighting systems. It enables the addition of a standard DALI communication interface to lighting devices



operating as slave nodes on a DALI network, allowing them to respond to a central controller for functions such as switching, dimming, and scene control. The design includes ready-to-use software that handles DALI signalling, interprets commands, sends responses, and interfaces with the physical DALI bus. It adheres to DALI standards, separates protocol handling from low-level device control, manages timing for reliable communication, and incorporates error detection to support operation in electrically noisy environments.

Applications. LED drivers, ballasts, luminaires, smart lighting modules, and retrofit upgrades used in commercial, industrial, and building automation systems.

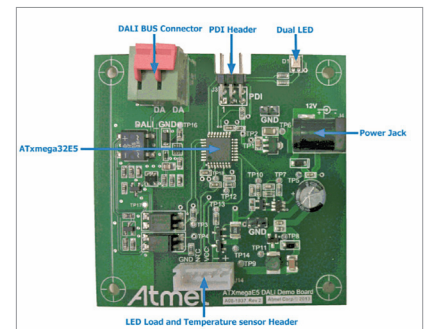
Link. <https://www.electronicsforu.com/electronics-projects/enabling-smarter-lighting-with-a-dali-slave-reference-design>

OEM Brand. STMicroelectronics

DALI Slave lighting control

In modern smart lighting and building automation systems, reliable lighting control depends on robust communication between the central controller and each luminaire. DALI is a widely used standard that enables this communication. A DALI Slave allows a lamp or LED driver to interpret controller commands and perform functions such as switching, dimming, and status reporting. The DALI Slave reference design

shows how to connect an AVR microcontroller to a DALI bus and control LED lights. It has three main parts: the DALI interface, the microcontroller, and the LED driver. The microcontroller decodes DALI commands, controls dimming, and drives the LEDs using its built-in timers and communication tools. The LED driver uses a buck converter with MOSFETs to provide smooth dimming up to 300mA and is placed on a separate board for easy testing and customisation.



Applications. Testing and customisation across a wide range of lighting applications.

Link. <https://www.electronicsforu.com/electronics-projects/reference-design-for-dali-slave-lighting-control>

OEM Brand. Microchip

These DALI reference designs make it easier for engineers to build LED lights that can be controlled reliably. By addressing core requirements such as DALI communication, dimming, isolation, and power management, you reduce the need to design every element from scratch. Whether you are building a basic LED driver, a complete DALI-2 lighting network, or a smart DALI Slave device, these reference designs help shorten development cycles, reduce errors, and support faster testing and customisation. **EFY**

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